

TensorFlow

- What is TensorFlow?
 - Features and Ecosystem Overview
 - Installing TensorFlow
 - TensorFlow vs Other Deep Learning Frameworks
 - Setting Up the Environment (Anaconda, Colab, Jupyter)
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2. Basics of TensorFlow

- Tensors:
 - What Are Tensors?
 - Creating and Manipulating Tensors
 - Tensor Operations and Broadcasting
 - Data Types and Shapes
 - Eager Execution vs Graph Execution
 - Working with GPU/TPU Acceleration
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3. TensorFlow Core API

- Creating and Executing Computational Graphs
 - Variables, Constants, and Placeholders
 - TensorFlow Functions (`tf.function`)
 - Using `tf.data` for Efficient Data Loading and Preprocessing
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4. Building Neural Networks with TensorFlow

4.1 Layers and Models

- Understanding TensorFlow Layers (`tf.keras.layers`)
- Sequential API
- Functional API
- Model Subclassing for Custom Architectures

4.2 Forward Propagation

- Creating Input, Hidden, and Output Layers
- Activation Functions (ReLU, Sigmoid, Tanh, Softmax)

4.3 Backpropagation

- Loss Functions:
 - Regression Losses (MSE, MAE)
 - Classification Losses (Cross-Entropy, Hinge)
 - Optimizers:
 - Gradient Descent
 - Adaptive Methods (Adam, RMSProp, AdaGrad)
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5. Data Handling in TensorFlow

- Using `tf.data.Dataset` for Large Datasets
 - Data Preprocessing Pipelines:
 - Normalization and Augmentation
 - Handling Image, Text, and Time-Series Data
 - Reading CSV, TFRecords, and HDF5 Files
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6. Advanced TensorFlow Features

- Saving and Loading Models (`SavedModel`, HDF5)
 - TensorFlow Callbacks:
 - Early Stopping
 - Model Checkpointing
 - TensorBoard for Visualization
 - Distributed Training:
 - Data Parallelism with `tf.distribute`
 - Training Across Multiple GPUs/TPUs
 - Mixed Precision Training for Speed Optimization
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7. Convolutional Neural Networks (CNNs)

- Implementing CNNs in TensorFlow:
 - Convolutional Layers

- Pooling Layers (Max, Average)
 - Flattening and Fully Connected Layers
 - Pretrained Models (MobileNet, ResNet, EfficientNet)
 - Transfer Learning:
 - Fine-Tuning
 - Feature Extraction
 - Applications:
 - Image Classification
 - Object Detection and Segmentation
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8. Recurrent Neural Networks (RNNs)

- Implementing RNNs in TensorFlow:
 - RNN Layers (`tf.keras.layers.SimpleRNN`)
 - Long Short-Term Memory (LSTM)
 - Gated Recurrent Units (GRU)
 - Time-Series Forecasting
 - Natural Language Processing (NLP):
 - Sentiment Analysis
 - Text Generation
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9. Transformers in TensorFlow

- Attention Mechanism
 - Building Transformers Using TensorFlow
 - Applications:
 - BERT and GPT for NLP
 - Vision Transformers (ViT)
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10. Generative Models

- Variational Autoencoders (VAEs):
 - Building Autoencoders with TensorFlow
 - Applications (Anomaly Detection, Data Compression)
- Generative Adversarial Networks (GANs):
 - Building GANs in TensorFlow
 - DCGAN, CycleGAN

- Applications:
 - Image Synthesis
 - Style Transfer
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11. Reinforcement Learning

- Basics of Reinforcement Learning
 - Implementing Deep Q-Learning in TensorFlow
 - Policy Gradient Methods
 - Applications (Games, Robotics)
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12. TensorFlow for Production

- Model Deployment:
 - TensorFlow Serving
 - TensorFlow.js for Web Applications
 - TensorFlow Lite for Mobile and Edge Devices
 - Exporting Models to ONNX
 - Optimizing Models for Deployment:
 - Quantization
 - Pruning
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13. Advanced TensorFlow Topics

- Custom Training Loops with `GradientTape`
 - Writing Custom Layers and Loss Functions
 - Working with Sparse Tensors
 - Debugging TensorFlow Models
 - Profiling and Optimization
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14. TensorFlow Ecosystem

- TensorFlow Hub for Pretrained Models
- TensorFlow Extended (TFX) for End-to-End ML Pipelines

- TensorFlow Probability for Probabilistic Models
 - TensorFlow Recommenders for Recommendation Systems
 - TensorFlow Graphics for 3D Data
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15. Visualization and Monitoring

- Using TensorBoard for Visualization:
 - Training Metrics
 - Graphs
 - Hyperparameter Tuning
 - Visualizing Feature Maps in CNNs
 - Grad-CAM for Model Interpretability
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16. Case Studies and Projects

Beginner Projects

- Handwritten Digit Classification (MNIST)
- Predicting House Prices with Dense Neural Networks

Intermediate Projects

- Image Classification with Transfer Learning
- Time-Series Forecasting (Stock Prices, Weather)

Advanced Projects

- Building a Chatbot with Transformers
 - Creating a GAN for Image Generation
 - Implementing an End-to-End ML Pipeline with TFX
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17. Research and Development

- Understanding Research Papers on TensorFlow
- Customizing TensorFlow for Novel Architectures
- Exploring New TensorFlow Libraries (e.g., KerasCV, TensorFlow Decision Forests)

